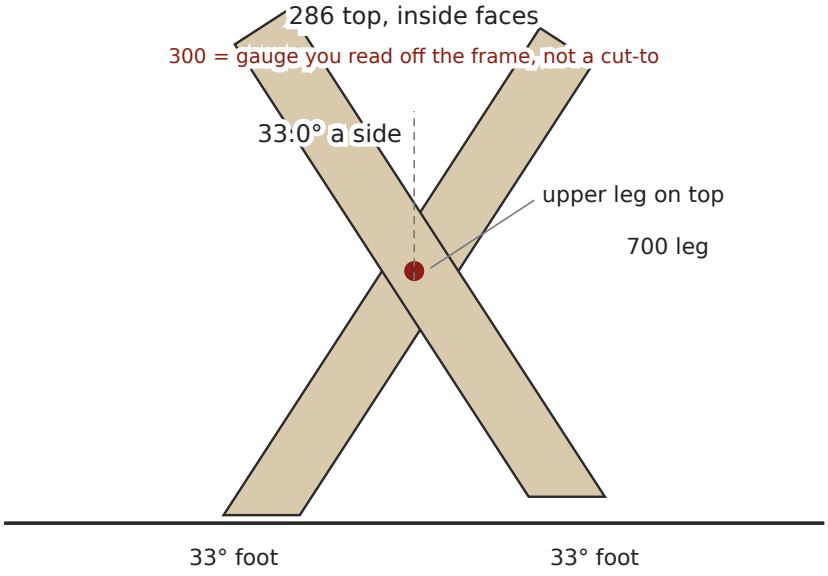
X-Leg Herb Trough

plan · v7

- Your numbers. Two X-leg frames carry a 900 long trough between them.
- Legs **4×700**, one end cut 33°. Cross planks **8×900**, four front and four back. Planks 80 wide × 23 thick.
- **The X is set by the top.** Both leg tops sit **300 apart, inside corner to inside corner** - that is your gauge. You **set the standing frame** so the top gauges 300, inside to inside, so every X matches.
- The two legs do not notch: the upper leg lies straight **on top of** the lower one, so there is an inside leg and an outside leg.
- The pieces that close the ends are not cut to a number. They **lie inside the V of the legs** and are **scribed off an offcut**, one on top of the last.
- **The one lengthwise trim.** The second floor plank lands one thickness high, so trim **one plank thickness** off its long edge. 23 mm is only the example for 23 mm stock - whatever the yard sells is the trim.
- Shown at 900 long, four planks deep. Dimensions in mm.

X JOINT - stage 1



Stage 1 - the joint that decides the box. Two 700 legs crossed at their centres, upper leg lying on the lower. The top span the 33° gives between the leg inner faces is **286** - worked out, not measured: 2 × (350·sin33° − 40÷cos33°) = 286. The **300** is the gauge you read off the standing frame, inside to inside - set it there, do not cut to it. Foot flush on the clamped plank and the top gauging 300: the frame stands true, and both frames match.

CUT LIST

PART	QTY	LENGTH	WIDTH	THICK	NOTE
Leg	4	700	80	23	one end cut 33°, all four one setting
Cross plank (wall)	8	900	80	23	4 front, 4 back
Side piece	6	scribed	80	23	3 per end, marked off an offcut
- of those, the bottom pair	2	900	80	23	first plank's inner long edge cut 22.5° into the V; second trimmed one plank thickness

BUY IT IN YOUR YARD'S STOCK

- **Timber: 6 lengths of 2 m, 80 wide × 23 thick.** Derivation, not a guess: the 4 legs (4×700 = 2,800) come out of 2 lengths, two legs per length; the 8 cross planks (8×900 = 7,200) out of 4 lengths, two per length with 200 off each.
- **The pinch point is the side pieces.** Six lengths leaves offcut of two 600s and four 200s. Scribe the six side pieces off the two 600s, longest first. Measure before you go back: if a course needs more than 600, buy a 7th length.
- **The trim is 23, one plank thickness.** Same rule as always, now with your number in it.
- **If you buy the 90 wide instead**, two numbers change and nothing else does: mark the crossing **45 across** (half of 90, not 40), and the top span between the leg inner faces becomes **274** (same formula, 45 in place of 40). 700, 900, 33°, 22.5°, the 23 trim: all the same. The drawing here is at 80 wide.
- Untreated pine or hardwood. Untreated: it is a food bed.
- **35 mm part-threaded screws** for the crossing and the side courses.
- A **6 mm or 10 mm bolt** through each crossing if you want it to never move.
- Stainless or coated only. Wood oil (Woodoc) outside and on the feet.

Two rules that decide this build

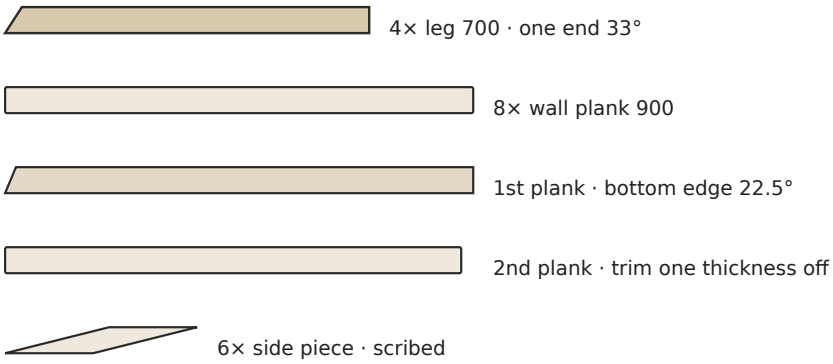
- **One saw setting for all four feet.** Cut every 33° end off the same stop. One foot cut on its own and the box stands on three legs.
- **Lock the crossing.** A single screw lets the X scissor. Add 3-4 screws, or bolt it.

Locked in from your build photos (14 Oct)

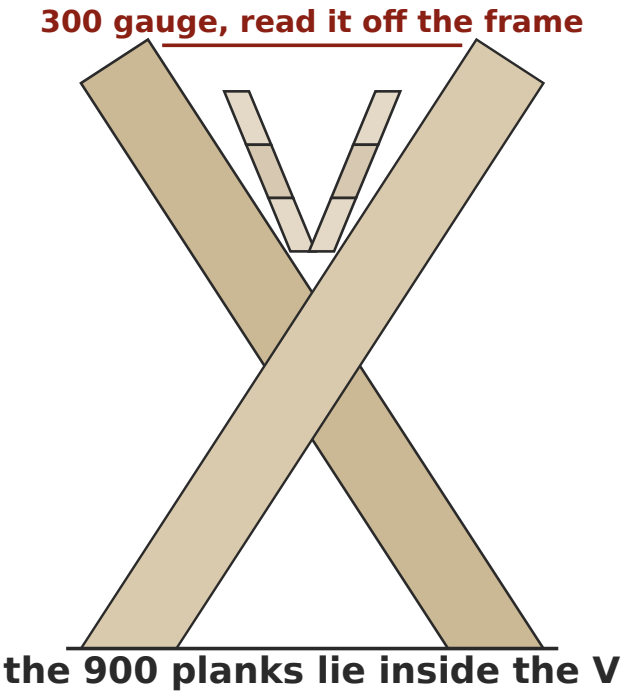
- **Top = 300, inside corner to inside corner.** Read it off the standing frame - it is the check, never a cut-to. Every X the same.
- **Feet cut at 33°** so they sit flat. Method: clamp a plank, leg flush, one centre screw into the middle of the plank.
- **The two legs cross one on top of the other.** No notch - the upper leg lies full-thickness on the lower, so there is an inside leg and an outside leg.
- **Spacer blocks at the shoulder.** Small blocks fill the gap between the angled leg head and the underside of the first planks. Drawn from the ~60 and ~75 mm marks in your shots.
- **One wall flush, one proud.** The back planks run **flush to the outside leg**; the front planks stand **one plank thickness proud** (23 mm for 23 mm stock), as in your shots.

Cut list, drawn what to cut

Every cut in the build. Two angles hold the whole thing: **33°** on the leg, **22.5°** in the V, and the rest square.



WHERE THE PLANKS SIT



End view: the pieces lie inside the V of the legs. The trough is not a tray sitting on top - it drops down inside the X, both long sides leaning in at 22.5° to a narrow bottom, and the leg tops stand proud above the rim. The lowest plank's bottom edge is cut at 22.5° so it sits level; the next lands one plank thickness higher, so trim one plank thickness off its long edge and the tops come level.

The two angles that fool everyone

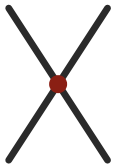
- 33° is the **leg** - the foot cut and the lean, one saw setting. Nothing else is 33°.
- 22.5° is the **V** - how far each side leans in, and the cut on the lowest plank's bottom edge to match it. A chosen call: it beds the plank into the V, it is **not** derived from the 33°.
- Each side above that is **scribed** off an offcut stood in the V, not set to a number.

The build in four stages

picture sequence

Four stages, in order. Do not start the sides until the walls are screwed down and level.

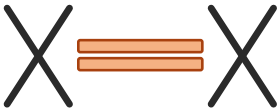
1



33° feet · screw the centre

Stage 1 - the X legs. Cut one end of each 700 leg at 33°. Cross two: the upper leg lies straight **on top of** the lower one. Centre 350 along, 40 across, one screw through the middle. Make two, mirrored - one set inside, one outside, both set to the same 300 top gauge.

2



first two in the V

Stage 2 - stand them and start the floor. Stand the two X sets parallel, set to the 300 top gauge. Cut one **long edge** of the first 900 at **22.5°** - the inner edge that lands on the far leg - so it beds into the V. The second lands one plank thickness higher, so trim **one plank thickness** off its long edge - 23 mm for 23 mm stock - so both tops come level.

3



4 front + 4 back

Stage 3 - the rest of the 900s. 4 planks on the front, 4 on the back, flush with the left and right X. The back sits one plank thickness proud (23 mm for this stock) behind the outside leg. Screw down front and back.

4



sides off an offcut

Stage 4 - the sides. Hold an offcut in the V, mark the angle, cut the first side piece. Next piece higher: mark off the first, cut, glue and screw on top. Third the same. Mirror the other end.

Stage by stage 1 – 2

#	STAGE	WHY
1	**The X legs.** • Four legs, 700 long. Cut **one end of each** at 33° - all four off the **same saw setting**. • Find the centre of each leg: 350 along, 40 across (half of 80). • Clamp a straight plank to the table. Press the 33° end flush against it. • Lay the second leg **straight on top of the first** to make the X. Pilot hole at the centre. • Drive a 35 mm part-threaded screw down through both at the centre so the upper leg sits on the lower one. • Then 3-4 more screws, or swap to a 6 mm or 10 mm bolt through the crossing. • Set the standing frame so the **top gauges 300, inside corner to inside corner** - read it, don't cut to it. Make the second set the same.	• One saw setting for all four feet = feet that sit level. Cut them one at a time and they fight. • The clamped plank is the reference - it stops the X leaning while you screw. • The legs do not notch into each other: one lies full-thickness on top of the other. That gives an inside leg and an outside leg. • A single screw lets the X scissor. Extra screws or a bolt lock the crossing. • 300 top gauge on both sets = the box sits level and the walls line up.
2	**Stand them and lay the first two planks.** • Stand the two X sets parallel, on the ground. • Lay the first two 900 planks in the V, one front, one back. • Cut **one long edge of the first plank at 22.5°** - the inner edge where it meets the far leg - so it beds into the V. • The second plank lands **one plank thickness** higher, so **trim one plank thickness off** its long edge - 23 mm for 23 mm stock - and both tops come level.	• Both X sets parallel = the box goes up straight. • The 22.5° on the inner edge is what lets the plank bed into the V instead of rocking on the leg corner. • Level first planks = level box. This is the one that sets it.

Stage by stage 3 – 4

#	STAGE	WHY
3	**Lay the rest of the 900s.** • Carry on stacking: 4 planks on the front, 4 on the back, one on top of the other. • All flush with the left and right X planks. • The back stacks one plank thickness proud (23 mm here), behind the outside leg. • Both sides finish at the same height. • Screw the stack down, front and back, into the X frames.	• Flush to the X = straight walls, no stepping. • The back offset clears the leg, so both walls meet the frame the same way. • Equal height both sides = a level rim.
4	**The sides, scribed off an offcut.** • Hold an offcut horizontal in the bottom of the V. Draw a line from inside to mark the angle of the first side piece. • Cut to the line, fit it in the V. • Hold a longer offcut above the first, mark off it, cut the second, glue and screw it on top of the first. • Third the same, on top of the second. • Do the same for the opposite end.	• The V is not a set angle you can measure off a drawing - you scribe each piece off the one below. • Glue **and** screw: end grain is weak, the glue takes the load, the screws hold it while it sets. • Three pieces fill the end from the floor to the rim.